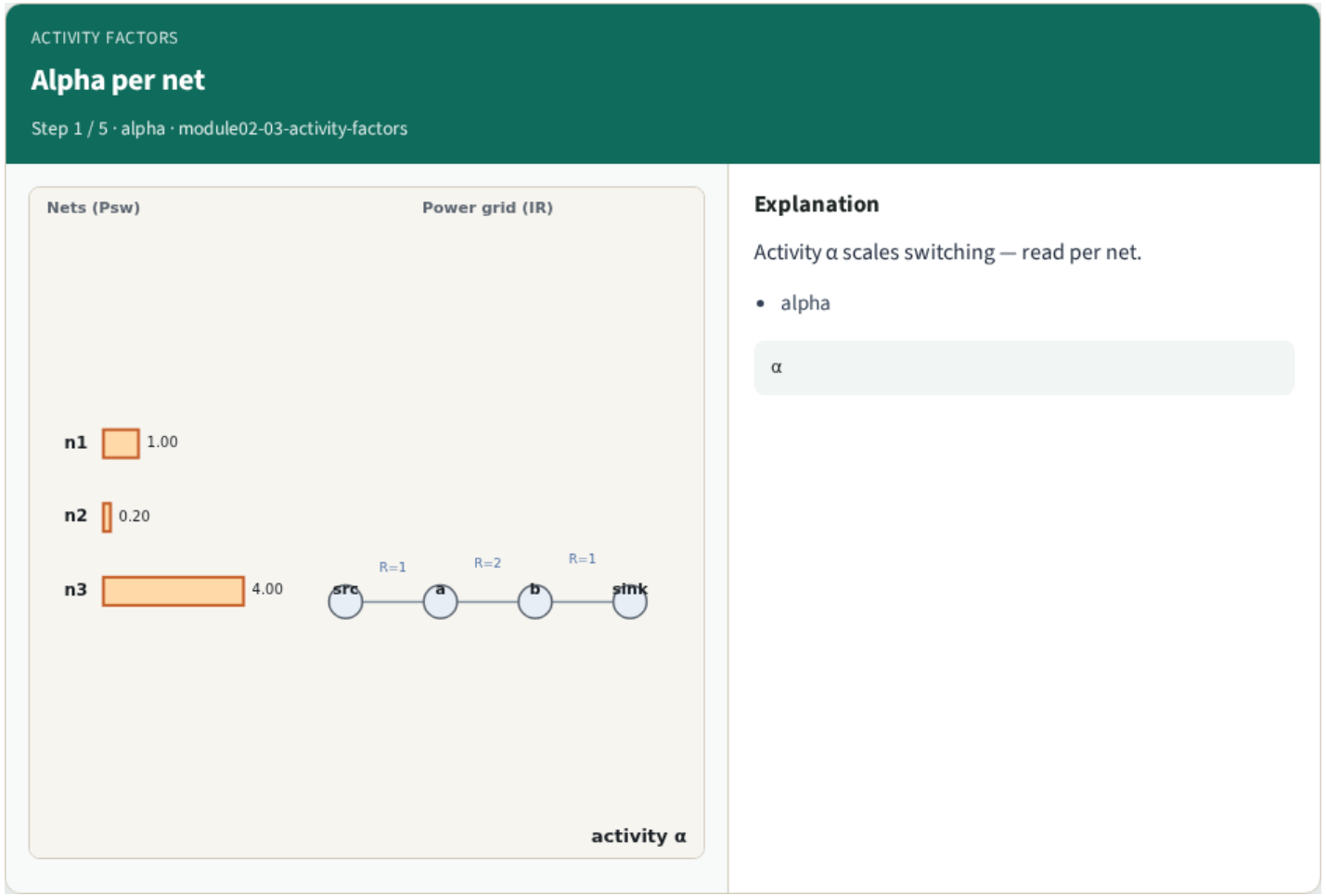
Why activity

Lower α on quiet nets reduces reported switching power

The idea

- GOLDEN: alphas n1=0.5, n2=0.2, n3=1.0

Alpha per net



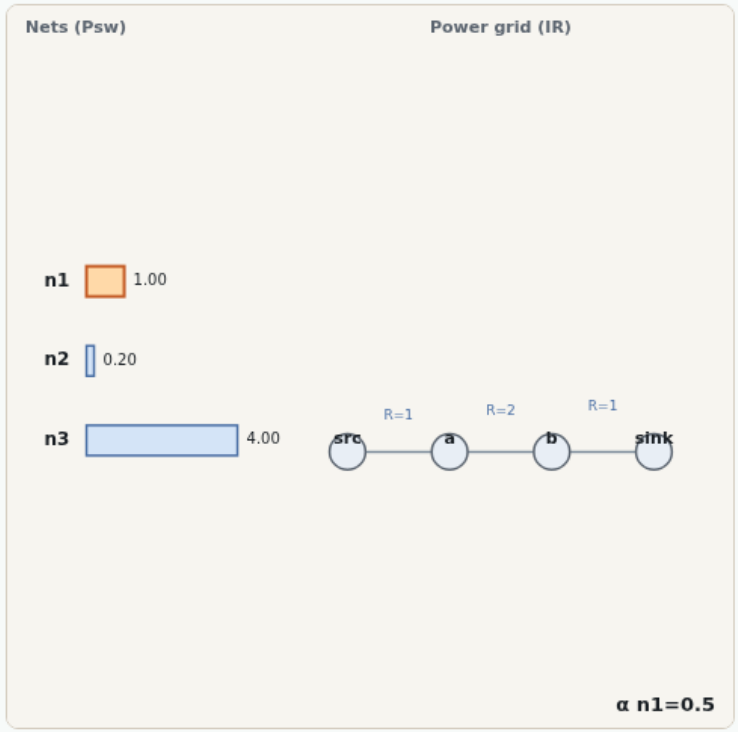
Alpha per net

n1 α

ACTIVITY FACTORS

n1 α

Step 2 / 5 · n1a · module02-03-activity-factors



Explanation

n1 α=0.5.

- n1

```
0.5
```

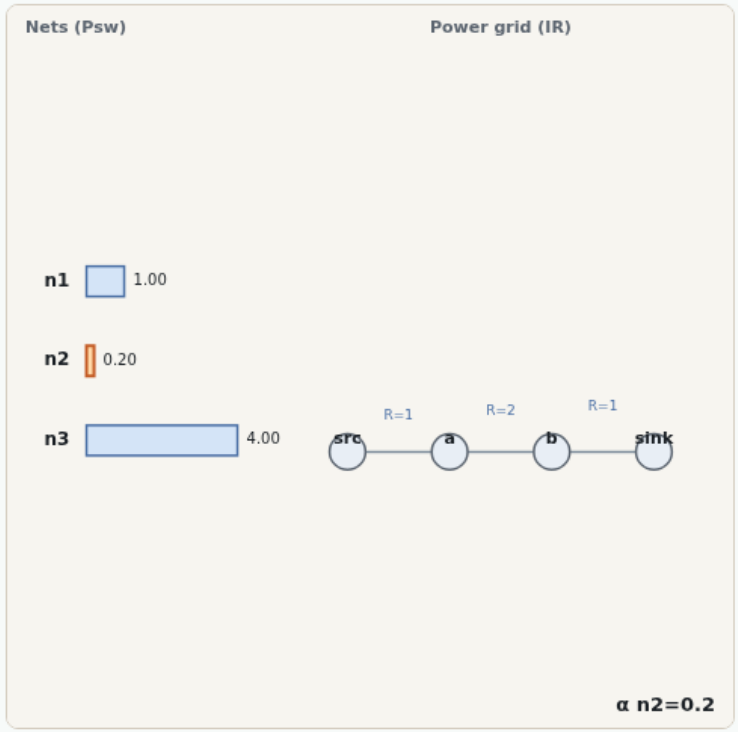
n1 α

n2 α

ACTIVITY FACTORS

n2 α

Step 3 / 5 · n2a · module02-03-activity-factors



Explanation

n2 α=0.2.

- n2

```
0.2
```

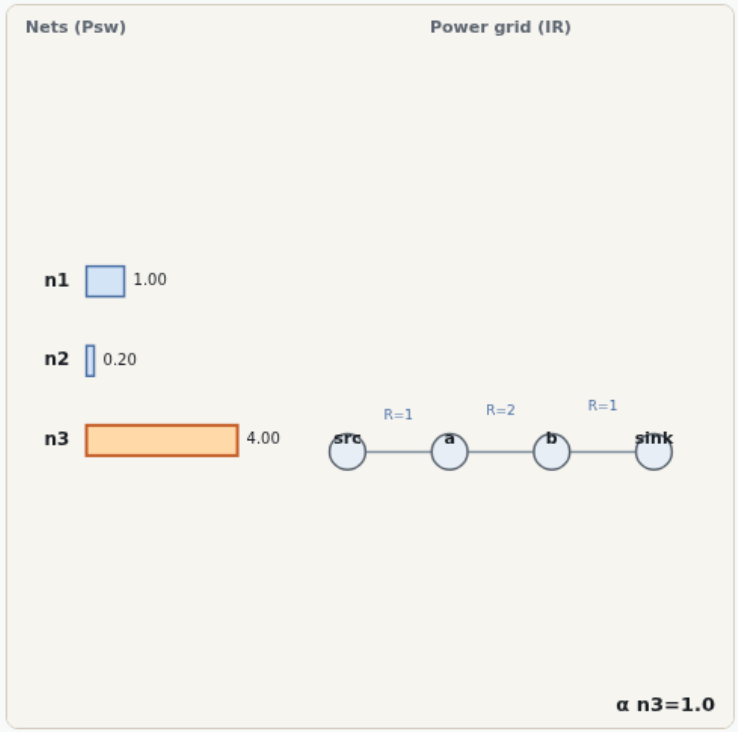
n2 α

n3 α

ACTIVITY FACTORS

n3 α

Step 4 / 5 · n3a · module02-03-activity-factors



Explanation

n3 α=1.0 (fully active).

- n3

```
1.0
```

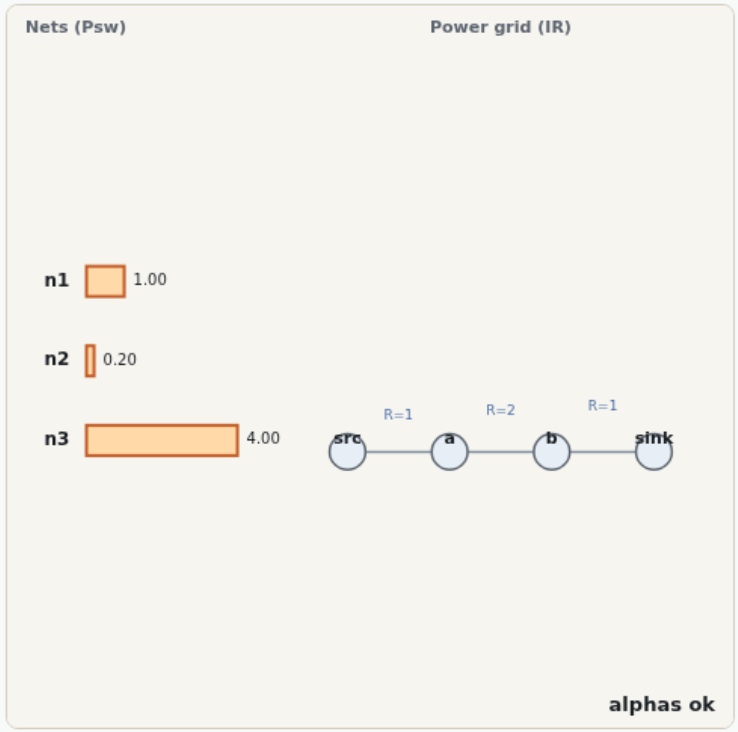
n3 α

Golden

ACTIVITY FACTORS

Golden

Step 5 / 5 · golden · module02-03-activity-factors



Explanation

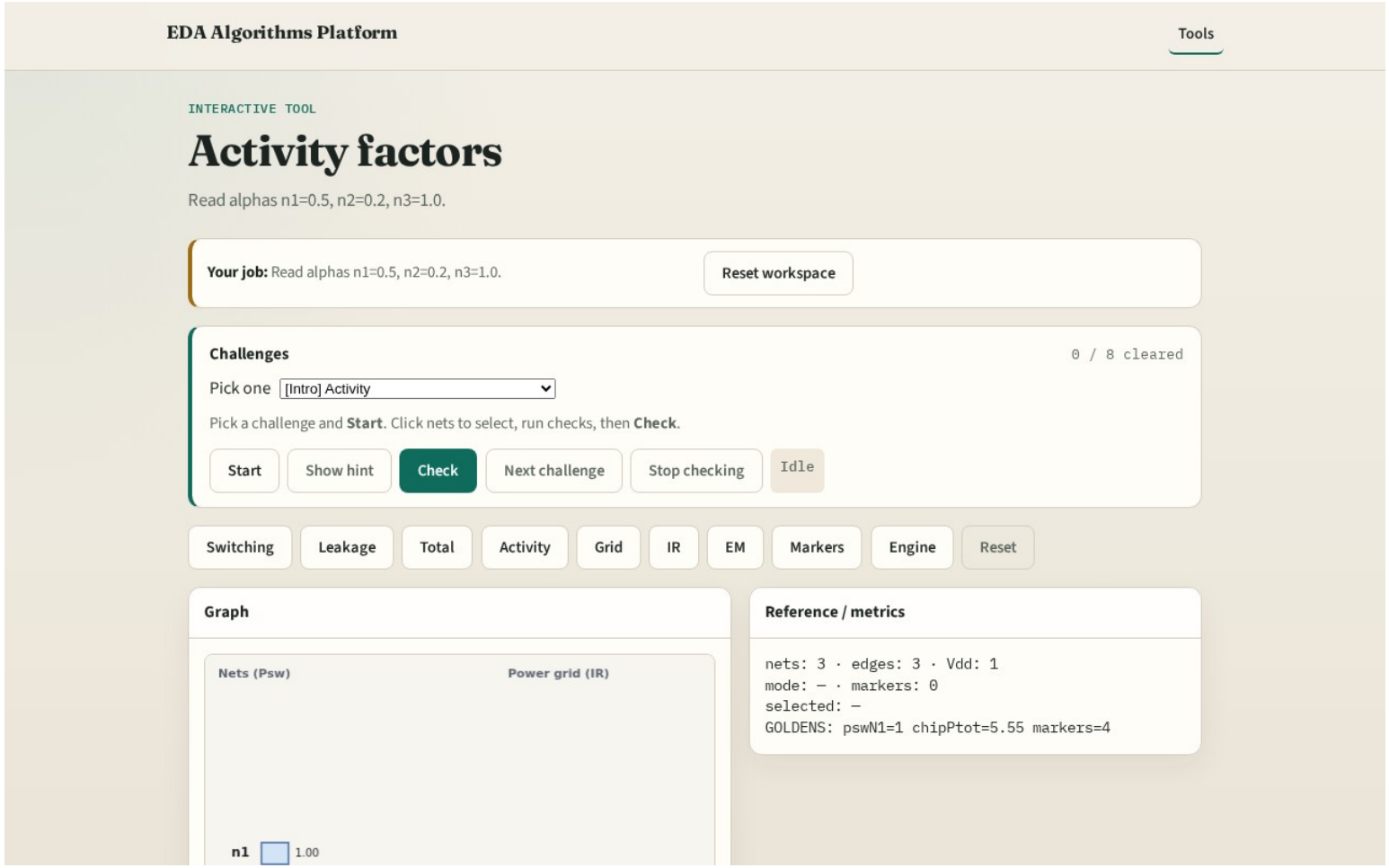
GOLDEN: alphas n1=0.5, n2=0.2, n3=1.0.

- GOLDEN

```
0.5/0.2/1.0
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call activity_factors(); assert alphas match goldens

Pitfalls

- Confusing α with Ileak
- Using mean instead of per-net

Your turn

- Next: power grid